

# Lecture 3 Tutorial — Data Distributions

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## Q1. Key Terms (Fill in the blanks)

Complete the statements.

1. \_\_\_\_\_ probability refers to the probability of an event given that another event has occurred.
2. \_\_\_\_\_ probability refers to the probability that two events occur together.
3. A probability distribution can be used as a \_\_\_\_\_ of uncertainty.

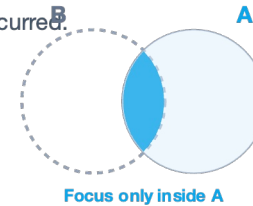
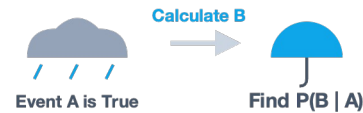
Q1

1

## Conditional Probability

**Definition:** The probability of an event given that another event has occurred.

**Example:** If it is raining (Event A),  
what is the chance I take an umbrella (Event B)?

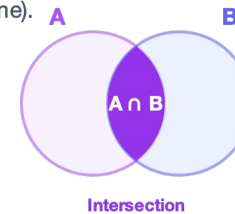


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## Joint Probability

**Definition:** The probability that two events occur together (at the same time).

**Example:** The chance of rolling a 6 on a die (A)  
AND flipping heads on a coin (B) together.

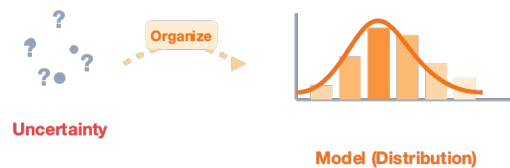


3

## Probability Distribution as a model

**Definition:** A distribution is a mathematical tool that describes and represents uncertainty.

**Example:** Flipping a coin many times turns random results  
into a clear expected chart.



## Q1 - Solution

### Q1. Key Terms (Fill in the blanks)

Complete the statements.

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1. Conditional probability 2. Joint probability 3. A model of uncertainty

## Q2. Variable Type (Multiple Choice)

Choose the correct option.

1. The number of successes in  $n$  trials is typically a:
  - A. Continuous variable
  - B. Discrete variable
  - C. Categorical variable
  - D. Joint variable
2. A measurement such as height or time is typically a:
  - A. Discrete variable
  - B. Continuous variable
  - C. Binomial variable
  - D. Poisson variable

## Q2- Solution

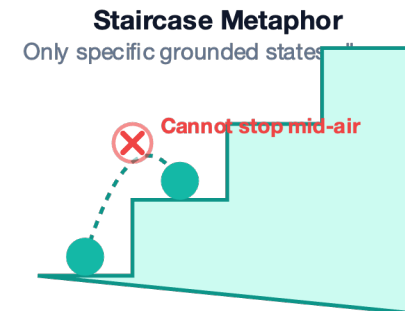
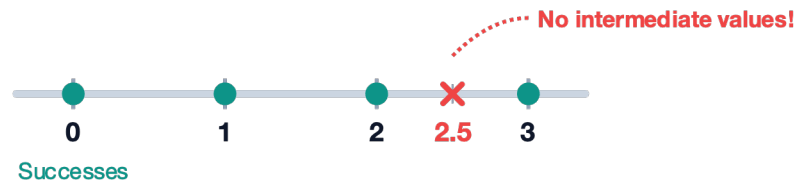
1

### Discrete Variable

**Core Concept:** Counts only distinct whole numbers (0, 1, 2, 3...).

**Question Context:** The number of successes in n trials.

**You CANNOT have 2.5 successes!**



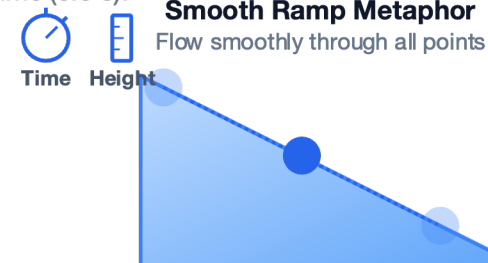
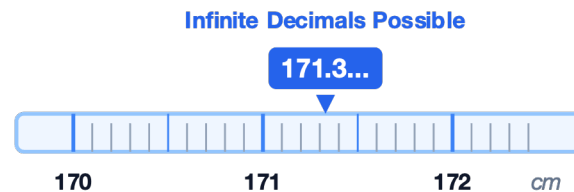
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### Continuous Variable

**Core Concept:** Can take *any numerical value* within a range.

**Question Context:** Measurements like Height (172.3 cm) or Time (5.6 s).

**These values flow continuously and include all decimals.**



### **Q3. True / False (Probability Reasoning)**

Write T (True) or F (False).

1. A low-probability event is impossible.
2. A probability distribution helps describe uncertainty.

## Q3 - Solution

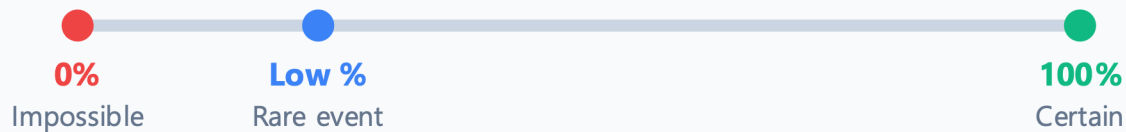
### Q3: PROBABILITY ASSESSMENT

#### 1. "A low-probability event is impossible."

FALSE

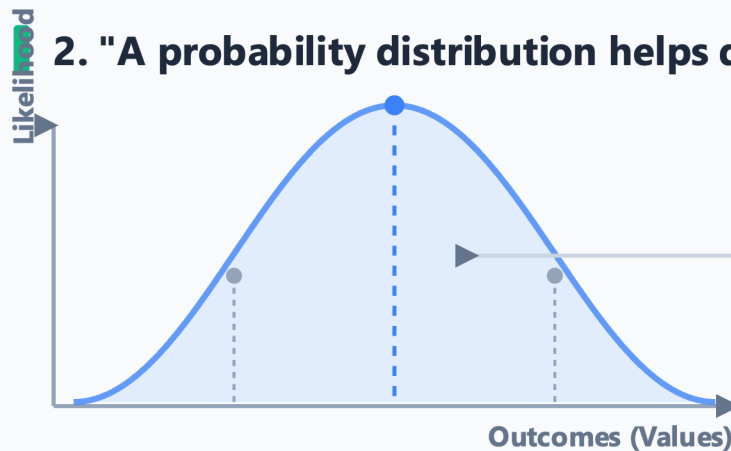
The Probability Scale

**Low  $\neq$  Zero.** Winning the lottery is extremely rare (low %), but it is NOT impossible (0%).



#### 2. "A probability distribution helps describe uncertainty."

TRUE



##### Mapping Uncertainty

A distribution is a mathematical map. It precisely plots the likelihood of every possible outcome occurring.



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## Q4. Binomial Distribution Parameters (Matching)

Match each symbol to its meaning.

**Symbols:**

A.  $n$

B.  $x$

C.  $p$

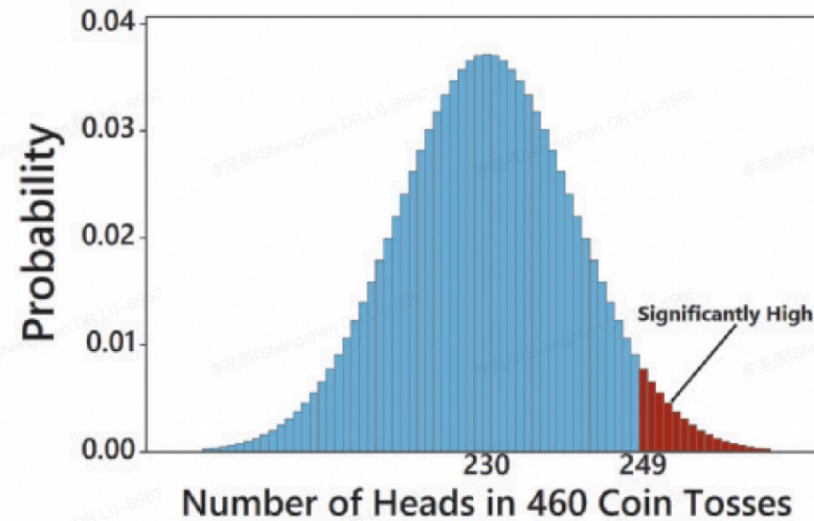
D.  $q$

**Meanings:**

1. Probability of failure in any one trial
2. Number of successes among trials
3. Number of trials
4. Probability of success in any one trial

Q4

- Significant Value



## Binomial Probability Distribution

$$P(x) = \frac{n!}{(n-x)!x!} \cdot p^x \cdot q^{n-x}$$

- $n \rightarrow$  number of trials
- $x \rightarrow$  number of successes among  $n$  trials
- $p \rightarrow$  probability of success in any one trial
- $q \rightarrow$  probability of failure in any one trial

Q4

## ANATOMY OF A BINOMIAL DISTRIBUTION

The 4 Core Parameters:  $n$ ,  $p$ ,  $q$ ,  $x$

### STEP 1: RULES FOR A SINGLE TRIAL

ONE SINGLE  
ATTEMPT

⚠ **EXAM TRAP:**

"Success" safely just means  
the outcome we are counting.

**$p$**

= Probability of **SUCCESS** in ONE trial

The underlying fixed chance of the target event happening.

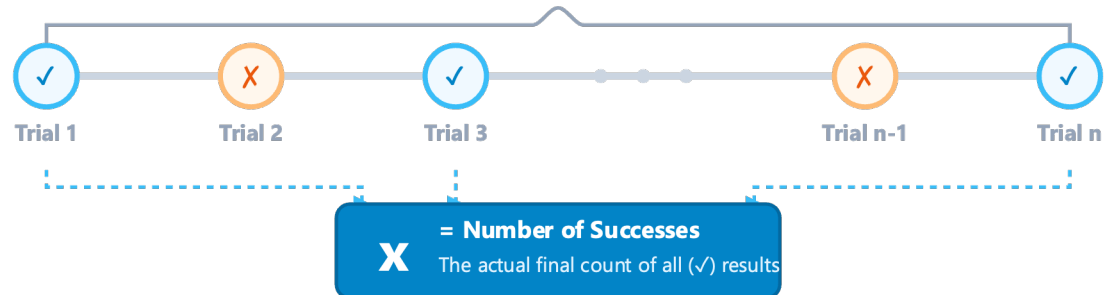
**$q$**

= Probability of **FAILURE** in ONE trial

The leftover opposing probability. (Formula:  $q = 1 - p$ )

### STEP 2: RUNNING THE FULL EXPERIMENT

**$n$  = Total Number of Independent Trials (Fixed Amount)**



## Q5. Choosing a Distribution

### **Question**

1. Number of defective items in 20 items (defective / not defective)
2. Number of calls received in one hour
3. Bell-shaped measurement noise around a mean

Q5

## 1. BINOMIAL

DISCRETE (Counts)

Key: "Pass / Fail"

Fixed Trials ( $n = 20$ )



Defective!

WHEN TO USE:

- Fixed number of trials
- Only TWO outcomes (Pass or Fail)

Scenario: Number of defective items in 20 items.

## 2. POISSON

DISCRETE (Counts)

Key: "Rate / Hour"

Random Events over Time



1 Hour Window

WHEN TO USE:

- Counting random events
- Within a fixed time or space

Scenario: Number of calls received in one hour.

## 3. NORMAL

(Gaussian)

CONTINUOUS (Decimals)

Key: "Symmetric / Bell"

Symmetric Clustering



$\mu$  (Mean)

WHEN TO USE:

- Continuous variables (e.g., Decimals)
- Symmetric, bell-shaped noise

Scenario: Measurement noise around a mean.

## Q5 - Solution

| Scenario                      | Best Distribution | Key Characteristics                                      |
|-------------------------------|-------------------|----------------------------------------------------------|
| 20 items, count defective     | Binomial          | Fixed trials ( $n=20$ ), two outcomes, count successes.  |
| Calls in 1 hour               | Poisson           | Count events in fixed time interval, known average rate. |
| Bell-shaped measurement noise | Normal            | Continuous, symmetric, bell curve around a mean.         |

### Question

1. Number of defective items in 20 items (defective / not defective)
2. Number of calls received in one hour
3. Bell-shaped measurement noise around a mean

### Answers

1. A (Binomial)
2. B (Poisson)
3. C (Normal/Gaussian)

## Q6 Short Response (Binomial vs Poisson)

### **Question**

1. What does  $\mu$  represent in the Poisson distribution?
2. State one key difference between  $p$  (Binomial) and  $\mu$  (Poisson).

## Q6

What does  $\mu$  represent in the Poisson distribution?

- $\mu$  (**mu**) = the **average number of events** that happen in a fixed time/area/interval.
- Example: On average, 5 cars pass a bridge per minute.
- Then  $\mu = 5$ .

In Binomial:

- $p$  = probability of success in **one trial**
- $\rightarrow$  It is always **between 0 and 1**.

In Poisson:

- $\mu$  = average number of events
- $\rightarrow$  It can be **any positive number** (0.5, 2, 5, 10, etc.), **can be greater than 1**.



## Q6 - Solution

### Question

1. What does  $\mu$  represent in the Poisson distribution?
2. State one key difference between  $p$  (Binomial) and  $\mu$  (Poisson).

### Answers

1.  $\mu$  is the **mean (average) number of events in a fixed interval**.
2.  $p$  is a probability (0 to 1);  $\mu$  is an average count (can be any positive number).

Q7 Normal Distribution (Fill in the blanks)

**Question**

1. The normal distribution is also called the \_\_\_\_\_ distribution.
2. The curve is \_\_\_\_\_-shaped and symmetric.
3.  $\mu$  denotes the \_\_\_\_\_,  $\sigma$  denotes the \_\_\_\_\_.

### BLANK 1: THE NAME

The normal distribution  
is also called the **GAUSSIAN** dist.

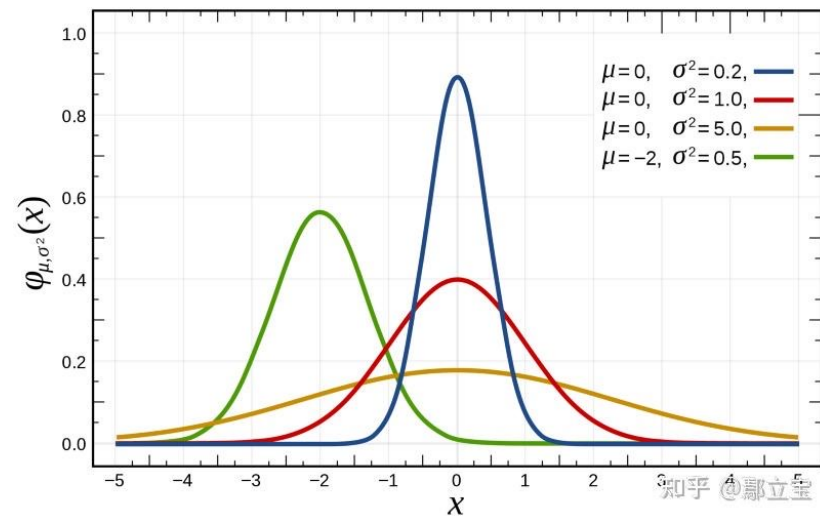
### BLANK 2: THE SHAPE

The curve is **BELL** -shaped  
and symmetric.

### BLANK 3: PARAMETERS

$\mu$  denotes the **MEAN**,

$\sigma$  denotes the **STANDARD DEVIATION**



Q7. Normal Distribution (Fill in the blanks)

**Question**

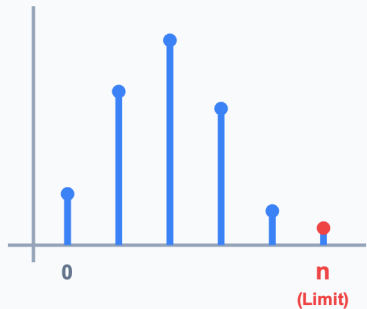
1. The normal distribution is also called the \_\_\_\_\_ distribution.
2. The curve is \_\_\_\_\_-shaped and symmetric.
3.  $\mu$  denotes the \_\_\_\_\_,  $\sigma$  denotes the \_\_\_\_\_.

**Answers**

1. Gaussian
2. bell
3. mean; standard deviation

## 1. BINOMIAL

DISCRETE (Counts)



### PARAMETERS:

$n$  = Fixed number of trials

$p$  = Probability of success

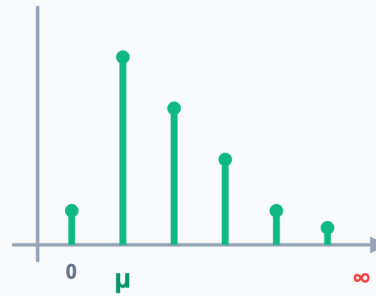
### DETAILED EXPLANATION:

Used when there are a **fixed number of trials**, and **only two outcomes**.

(Examples: success/failure, defective/normal)

## 2. POISSON

DISCRETE (Counts)



### PARAMETERS:

$\mu$  = Average number of events  
(Sometimes written as  $\lambda$  lambda)

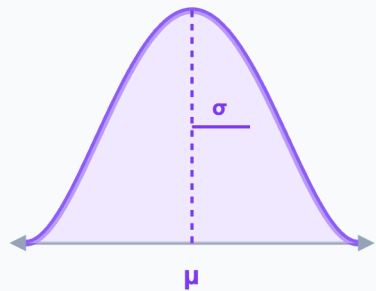
### DETAILED EXPLANATION:

Used for counting the **number of events** in a **fixed time or space**.

(Examples: calls per hour, customers per minute)

## 3. NORMAL (Gaussian)

CONTINUOUS (Data)



### PARAMETERS:

$\mu$  = Mean (The exact center)

$\sigma$  = Standard Deviation (Spread)

### DETAILED EXPLANATION:

Used for **continuous, symmetric, bell-shaped** data.

(Examples: measurements, errors, noise)

| Feature                      |  BINOMIAL                                                                                         |  POISSON                                                                                    |  NORMAL (Gaussian)                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Type                    | <b>Discrete</b><br>(Counts / Whole numbers)                                                                                                                                        | <b>Discrete</b><br>(Counts / Whole numbers)                                                                                                                                    | <b>Continuous</b><br>(Measurements / Decimals allowed)                                                                                                         |
| Key Parameters               | <b>n</b> = Total number of trials<br><b>p</b> = Probability of success                                                                                                             | <b><math>\mu</math></b> (or <b><math>\lambda</math></b> ) = Average rate of events                                                                                             | <b><math>\mu</math></b> = Mean (center)<br><b><math>\sigma</math></b> = Standard deviation (spread)                                                            |
| Boundaries (Limits)          |  <b>Fixed Upper Limit</b><br>(Count can never exceed $n$ )                                        |  <b>No Upper Limit</b><br>(Counts can go to infinity)                                       | <b><math>\leftrightarrow</math> No Boundaries</b><br>(Extends infinitely in both directions)                                                                   |
| When to Use (Core condition) | 1. There is a <b>fixed number</b> of independent trials.<br>2. There are <b>only TWO</b> possible outcomes (Pass/Fail).                                                            | Counting the number of random events happening within a <b>fixed interval of time or space</b> .                                                                               | Modeling <b>continuous, symmetric</b> data that naturally clusters around a central average.                                                                   |
| Visual Graph Shape           | Stem plot that completely stops at $x=n$                                                                                                                                           | Stem plot with a long tail trailing off to the right                                                                                                                           | Perfectly smooth, symmetric <b>Bell-shaped curve</b>                                                                                                           |
| Classic Exam Examples        | <ul style="list-style-type: none"> <li>Number of <b>defective items in 20 items</b></li> <li>Number of Heads in 10 coin flips</li> <li>Number of replies from 50 emails</li> </ul> | <ul style="list-style-type: none"> <li>Number of <b>calls received in one hour</b></li> <li>Customers arriving per minute</li> <li>Number of typos on a single page</li> </ul> | <ul style="list-style-type: none"> <li><b>Measurement noise</b> around a mean</li> <li>Manufacturing size errors</li> <li>Human heights and weights</li> </ul> |

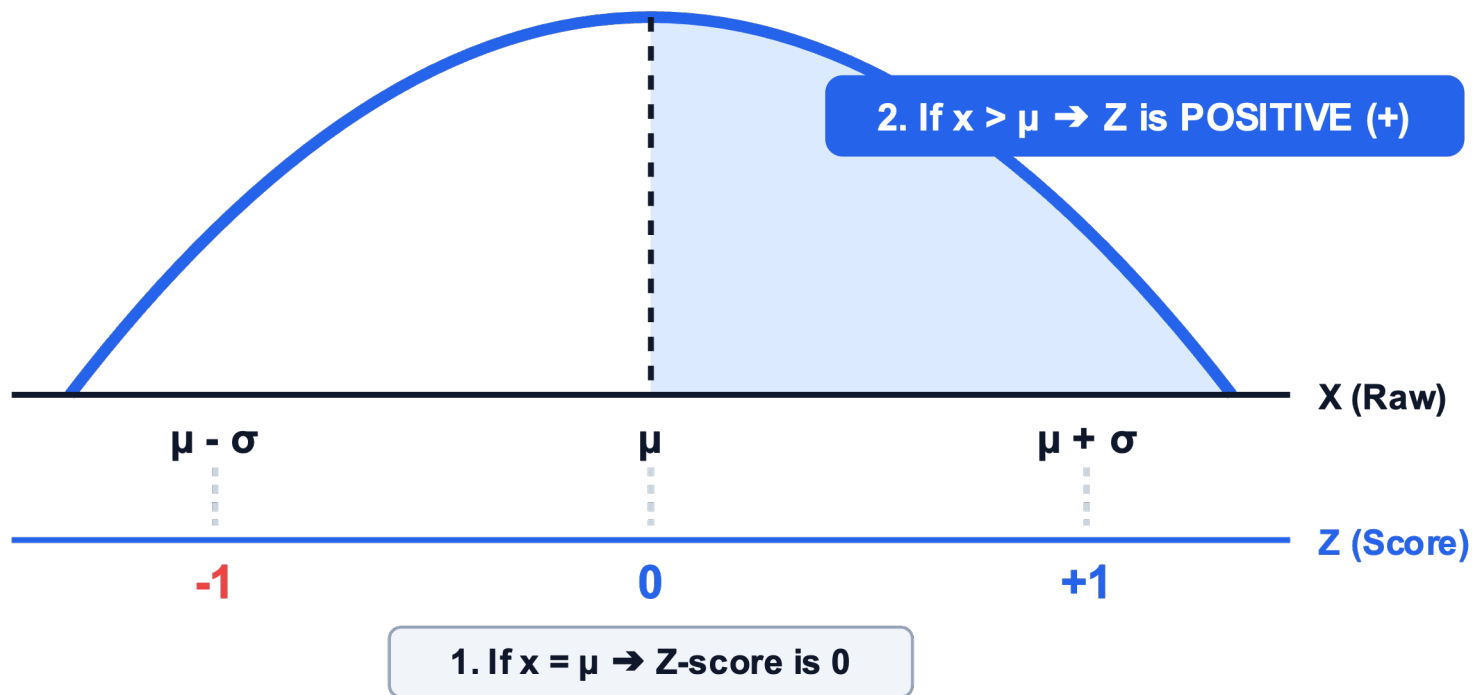
### Q8. Z Score

#### **Question**

Complete the formula:  $z = (x - \text{_____}) / \text{_____}$

1. If  $x > \mu$ , is  $z$  positive, negative, or zero?
2. If  $x = \mu$ , what is  $z$ ?

## Q8. Z-SCORE: $z = (x - \mu) / \sigma$





Q8. Z Score

**Question**

Complete the formula:  $z = (x - \text{_____}) / \text{_____}$

1. If  $x > \mu$ , is  $z$  positive, negative, or zero?
2. If  $x = \mu$ , what is  $z$ ?

**Answers**

$$z = (x - \mu) / \sigma$$

1. Positive
2. 0

### Q9. Central Limit Theorem (CLT)

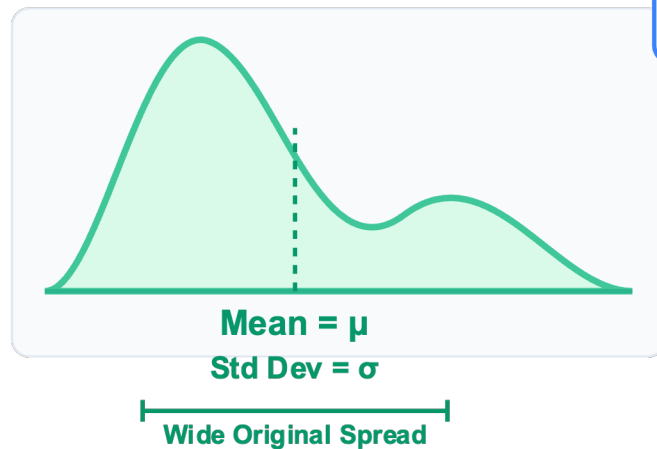
#### **Question**

For samples of the same size  $n$ , with  $n > \underline{\hspace{1cm}}$ , the sampling distribution of  $\bar{x}$  can be approximated by a normal distribution with mean  $\underline{\hspace{1cm}}$  and standard deviation  $\underline{\hspace{1cm}}$ .

## 1. ORIGINAL POPULATION

Shape: Literally ANYTHING!

(Skewed, bimodal, messy... it doesn't matter)



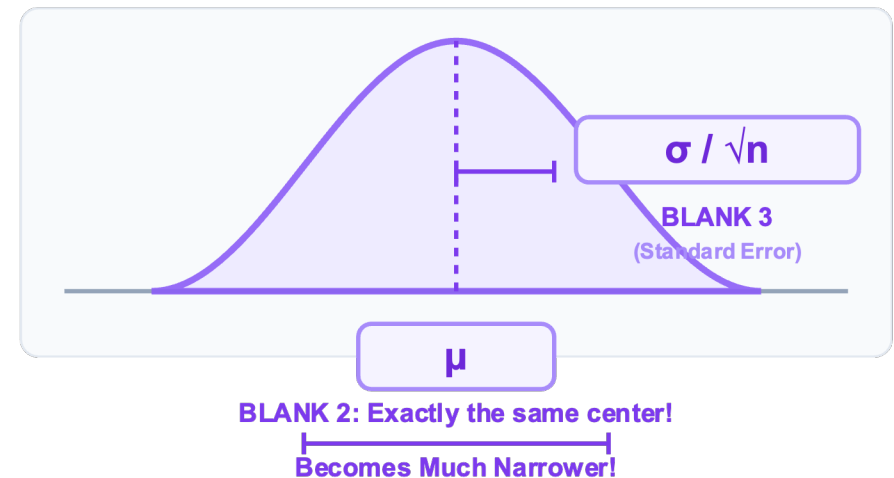
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 $n > 30$

Take samples of  
→  
Calculate  $\bar{x}$

## 2. SAMPLING DISTRIBUTION of $\bar{x}$

Shape: ALWAYS A PERFECT BELL CURVE!

(This is the "Magic" of the theorem)



### DETAILED EXPLANATION SUMMARY:

- When  $n > 30$ , the distribution of sample means ( $\bar{x}$ ) always follows a normal distribution, regardless of the original population's shape.
- The mean of these sample means is identical to the original population mean ( $\mu$ ).
- The standard deviation shrinks rapidly, becoming the "Standard Error", calculated as  $(\sigma / \sqrt{n})$ .

### Q9. Central Limit Theorem (CLT)

#### **Question**

For samples of the same size  $n$ , with  $n > \underline{\hspace{1cm}}$ , the sampling distribution of  $\bar{x}$  can be approximated by a normal distribution with mean  $\underline{\hspace{1cm}}$  and standard deviation  $\underline{\hspace{1cm}}$ .

#### **Answers**

30;  $\mu$ ;  $\sigma/\sqrt{n}$

Q10. CLT True / False

**Question**

1. The population must be normally distributed for CLT.
2. Simple random samples are required.
3. Samples should be the same size.
4. Population uses mean  $\mu$  and standard deviation  $\sigma$ .

Q10. CLT True / False

**Question**

1. The population must be normally distributed for CLT.
2. Simple random samples are required.
3. Samples should be the same size.
4. Population uses mean  $\mu$  and standard deviation  $\sigma$ .

1. **False:** CLT works for *any* population shape when  $n$  is large enough.
2. **True:** CLT requires **simple random sampling** to be valid.
3. **True:** All samples must have the **same size  $n$** .
4. **True:** Population is described by  $\mu$  (**mean**) and  $\sigma$  (**standard deviation**).

### Q11. Covariance Matrix and GMM

#### Question

1. GMM is used for data distributed in a \_\_\_\_\_ way.
2. Covariance matrix describes:  
A. Mean only B. Variances and covariances C. Conditional probability D. Expected value
3. Sum of mixture weights  $\pi_i = ?$

## Q11. Covariance Matrix and GMM

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### GMM Application

Q: GMM is used for data distributed in a \_\_\_\_\_ way.

**clustered**

Assumes data comes from a mix of multiple Gaussian distributions, ideal for clustered data.

### Covariance Matrix

Q: What does it describe?

A. Mean only B. Variances/Covariances  
C. Probability D. Expected value

**Answer: B**

Contains variances (diagonal) and covariances (off-diagonal), quantifying how variables vary together.

### Mixture Weights

Q: What is the sum of mixture weights  $\pi_i$  ?

**Sum = 1 (Unity)**

Weights represent probabilities of each component, so they must form a valid probability distribution.



## Q11. Detailed Explanation: Student Performance Analysis



### Scenario: Grouping Students

Goal: Given math and English scores, automatically cluster students with similar performance (identifying different ability groups).



### GMM Application ( $K=3$ )

Model splits students into 3 clusters: **High (90+)**, **Average (70-85)**, **Low (<60)**. The overall distribution is a mixture of these three normal distributions.



### Covariance Matrix

Determines shape/direction.

**Positive covariance** means high math score implies high English score. Visually, data clouds are tilted ellipses.



### Mixture Weights ( $\pi_k$ )

Proportions of each subgroup: **20% High, 60% Average, 20% Low**. Weights must sum to 100% (1.0).



### Key Takeaways for GMM Model

- 1. Number of Groups & Centers ( $\mu$ ):** How many clusters? Where are their means?
- 2. Shape & Correlation ( $\Sigma$ ):** Are variables correlated? What's the shape of the data cloud?
- 3. Proportions ( $\pi$ ):** What percentage of the total does each group occupy?

## Q12. Model Fitting and EM

### Question

1. Write one question to evaluate model fitness.
2. Expand EM: E = \_\_\_\_\_, M = \_\_\_\_\_

## Q12. Model Fitting and EM

### **Question**

1. Write one question to evaluate model fitness.
2. Expand EM: E = \_\_\_\_\_, M = \_\_\_\_\_

### **Answers**

1. Example: Does the model fit the observed data well?
2. Expectation; Maximisation

## Q12. Model Fitness & EM Algorithm ( Extended Section )

### Evaluate Model Fitness

**Q:** How to evaluate model fitness?

**Standard:** "What is the log-likelihood of observed data given current parameters?" (Higher = Better)  
Does the model fit the observed data well?

**Algorithmic:** "Has the log-likelihood converged (stopped increasing)?" (Convergence = Optimal)

### Expand EM Algorithm

**Q:** What does EM stand for?

**E:** Expectation (期望)

**M:** Maximization (最大化)



**Key Takeaway:** High log-likelihood and convergence confirm a good model fit. The EM algorithm iteratively applies Expectation and Maximization steps to find optimal parameters.

# The “Chicken-and-Egg” Problem in GMM ( Extended Section )



## Scenario: Unsupervised Learning

We have student scores (math & English) as a scatter plot, but no labels indicating performance levels.



## The Dilemma

- **If we knew labels:** We could easily calculate model parameters (mean, covariance).
- **If we knew parameters:** We could easily assign labels to students.



## The Reality (Chicken-and-Egg)

We know **neither** the labels nor the parameters. This deadlock is what the EM algorithm is designed to break.

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# EM Algorithm Step-by-Step (with Student Example) ( Extended Section )



## Step 1: Initialization

### Action:

Randomly place 3 Gaussian distributions (circles) with arbitrary means.

### Result:

The initial model has very poor fitness to the data.



## Step 2: E-Step (Expectation)

### Action:

Calculate the probability each student belongs to each cluster (Soft Assignment).

### Example:

Student Li (98/95) has **99%** chance in "High" cluster.



## Step 3: M-Step (Maximization)

### Action:

Use probabilities as weights to recalculate cluster means/covariances.

### Goal:

Maximize the likelihood the model fits the data.



**Key Insight:** Iterate between E-step (soft assignment) and M-step (re-estimate parameters) until convergence. The model gradually improves with each cycle.

# EM Iteration & Convergence ( Extended Section )

## The Iterative Loop



**E-step (Expectation):** Recalculate probabilities using adjusted clusters.



**M-step (Maximization):** Adjust clusters based on new probabilities.

This E-M cycle repeats **iteratively**, feeding the output of one step into the next.

## Convergence Criteria



**Stability:** Model parameters (clusters) stop changing significantly.



**Log-Likelihood:** Likelihood stops increasing (plateaus).

The algorithm halts when these conditions are met, indicating an **optimal fit** to the data.



## Final Outcome

Upon convergence, the model successfully identifies the hidden student groups (latent variables) and finds the parameters that best explain the observed data.

# Evaluating Model Fitness: The Key Metric ( Extended Section )



**The Ultimate Question:** How does the algorithm know it's improving?



**The Answer:** By tracking the **log-likelihood** of the observed data given the current model parameters.



**Why it works:** After every M-step, the log-likelihood **monotonically increases** (or stays the same). It never decreases.



**Convergence Criterion:** When the log-likelihood plateaus (stops increasing), the model has achieved the best possible fit.